Formula

$$ (\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} $$

Example

Campaign response: difference = 0.1625, SE = $\sqrt{(0.60 \cdot 0.40/150 + 0.4375 \cdot 0.5625/160)}$ ⇒ 95% CI ≈ (0.05, 0.25).

Interpretation

Since 0 is outside interval, the difference is significant at $\alpha = 0.05$.